

Instructions to install 4-parameter model of modified recombination

A ready-to-run modified version of CAMB is included in the package. If you wish to use provided version directly, proceed as follows.

Option 1: Build the provided Modified CAMB

From the main CAMB directory:

```
$ python setup.py build
```

Option2: Install CAMB from scratch and apply the modifications

If you prefer to start from a fresh installation of CAMB, follow the steps below.

Install CAMB via:

```
$ cd /path/to/cobaya_packages/code  
$ git clone --recursive https://github.com/cmbant/CAMB.git
```

Under CAMB/fortran directory, replace the files `recfast.f90`, `model.f90`, `camb.f90` with the versions provided in the package.

Under CAMB/camb directory, replace `model.py` with the modified version included in the package.

After implementing these changes:

Under CAMB/fortran directory:

```
$ make clean  
$ make
```

Then return to the main CAMB directory and build the Python wrapper:

```
$ python setup.py build
```

Running with Cobaya

A sample YAML configuration file is provided:

```
"planck_plusmb_bao_4mod.yaml"
```

You might need to install additional Likelihoods on Cobaya such as natively implemented SN Pantheon+, and DESI BAO, or external Likelihoods such as Planck PR4 Lensing.

Model Description

A full description of the 4-parameter model is available in [arXiv:2504.15274](#) and [arXiv:2411.16678](#).

The corresponding parameter mappings are:

- `zref` : Reference redshift at which the Gaussian-shaped bump in the ionization fraction peaks.
- `amds` : Amplitude of the Gaussian modification to $x_e(z)$.
- `sigw` : Width of the Gaussian bump at half maximum.
- `dzin` : Overall redshift shift applied to the unmodified ionization history.

Fixing the Licence Error of CAMB (If Encountered)

If a licence-related error occurs when compiling CAMB, proceed as follows.

Under CAMB directory, find the file `pyproject.toml` and open it.

Locate the `[project]` section and look for these lines:

```
license = "LicenseRef-LGPLv3-arxiv"  
license-files = ["LICENSE"]
```

Replace them with:

```
license = {text = "LGPLv3"}
```